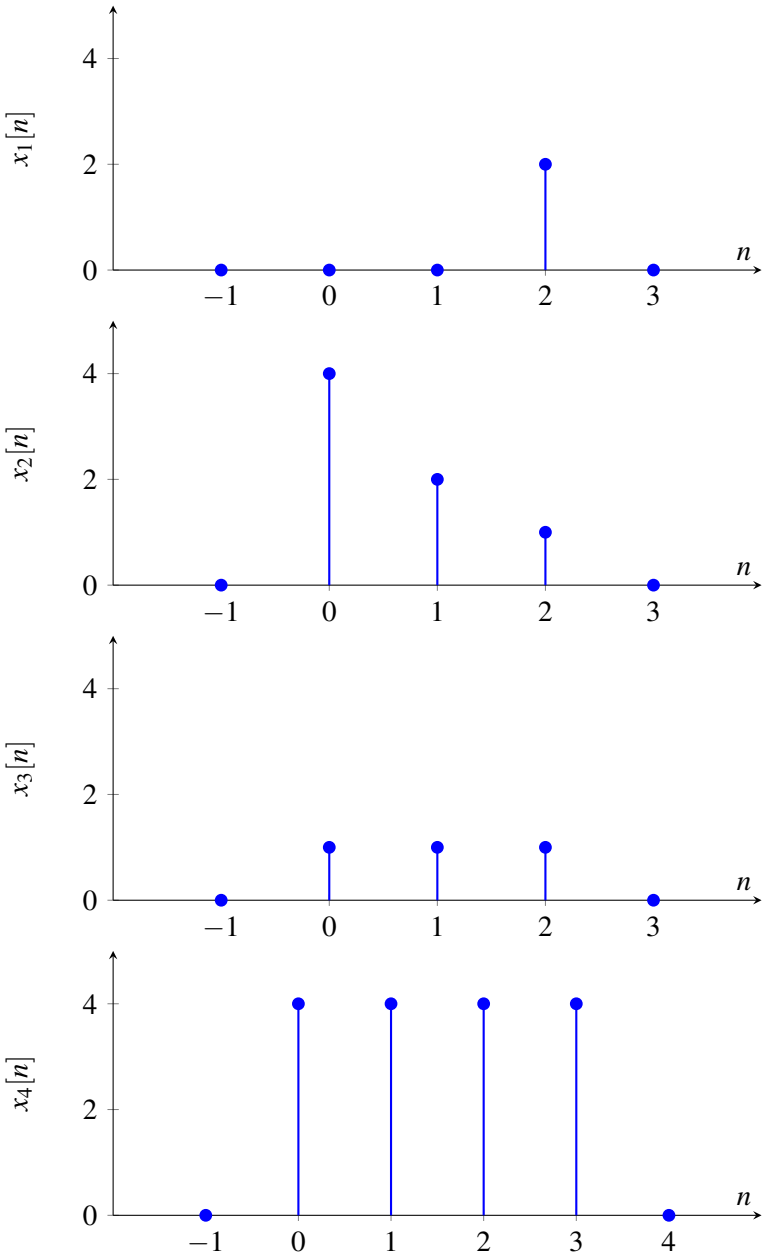
EE 66 Fall 2026 — Signals, Dynamics, and Information — Discussion 10A

1. DT Convolution Practice [Inspired by EE 120]

Consider the four discrete-time signals below.

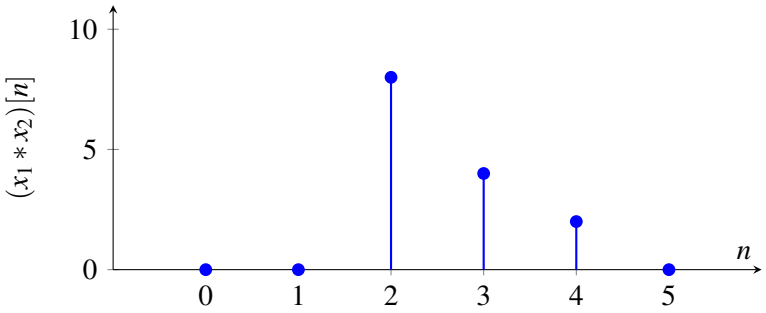


Carefully sketch and label plots of the following convoluted signals.

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(a) $(x_1 * x_2)[n]$

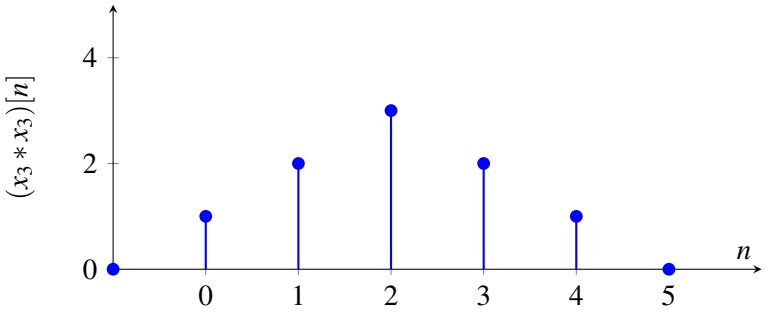
Answer:



Plot of $(x_1 * x_2)[n]$

(b) $(x_3 * x_3)[n]$

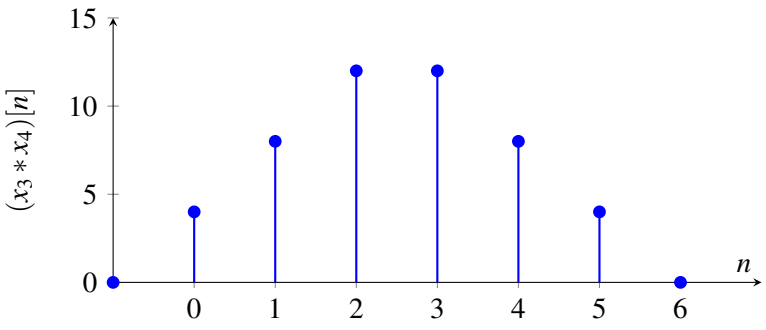
Answer:



Plot of $(x_3 * x_3)[n]$

(c) $(x_3 * x_4)[n]$

Answer:



Plot of $(x_3 * x_4)[n]$

2. Rolling Dice with Convolution

Let X and Y be the values of two independent rolls of a fair six-sided die. The probability mass function (PMF) of X assigns a probability to each integer:

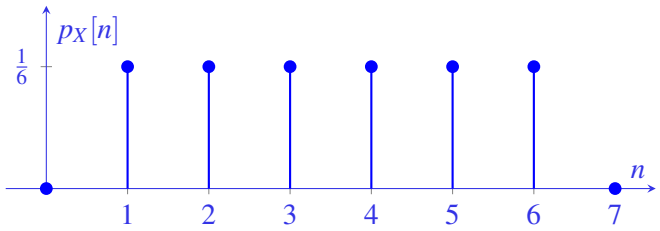
$$p_X[n] = \Pr(X = n) = \begin{cases} \frac{1}{6}, & n \in \{1,2,3,4,5,6\}, \\ 0, & \text{otherwise.} \end{cases}$$

Y has the same PMF, $p_Y[n] = p_X[n]$. Since the PMF is defined for every integer n, we can treat it as a discrete-time signal.

(a) Sketch $p_X[n]$ as a discrete-time signal. Write $p_X[n]$ as a sum of shifted impulses $\delta[n-k]$.

Answer: The signal is a rectangular pulse of height $\frac{1}{6}$ on $n = 1, \ldots, 6$:

$$p_X[n] = \frac{1}{6} \sum_{k=1}^{6} \delta[n-k].$$



(b) Let $Z = X + Y$. The event $\{Z = n\}$ happens exactly when $X = k$ and $Y = n - k$ for some integer k, and these events are disjoint for different values of k. Use this fact and the independence of X and Y to show that

$$p_Z[n] = \Pr(Z = n) = \sum_{k=-\infty}^{\infty} p_X[k]\, p_Y[n-k] = (p_X * p_Y)[n].$$

Answer: Since the events $\{X = k,\ Y = n-k\}$ are disjoint for different k, and together they make up the event $\{Z = n\}$, their probabilities add:

$$\Pr(Z = n) = \sum_{k=-\infty}^{\infty} \Pr(X = k,\ Y = n-k).$$

By independence, the joint probability factors as a product:

$$\Pr(X = k,\ Y = n-k) = \Pr(X = k)\Pr(Y = n-k) = p_X[k]\, p_Y[n-k].$$

Substituting this product gives

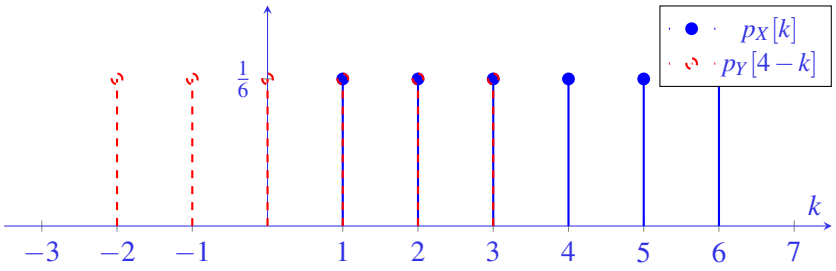
$$p_Z[n] = \sum_{k=-\infty}^{\infty} p_X[k]\, p_Y[n-k],$$

which is the convolution sum. The PMF of a sum of independent random variables is the convolution of their PMFs.

(c) Compute $p_Z[n]$ for all integers n by evaluating the convolution. Sketch $p_X[k]$ and $p_Y[n-k]$ as functions of k for $n=4$, and count the values of k where both are nonzero. Then sketch $p_Z[n]$.

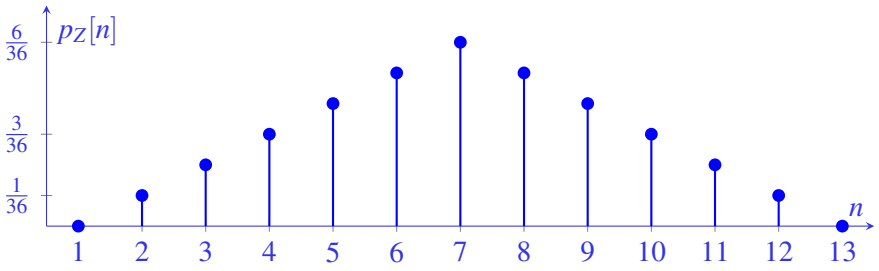
Answer: As a function of k, $p_Y[n-k]$ is nonzero for $1 \le n-k \le 6$, or $n-6 \le k \le n-1$. Each overlapping value of k contributes $\frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36}$, so $p_Z[n]$ equals $\frac{1}{36}$ times the number of values of k in both $\{1,\dots,6\}$ and $\{n-6,\dots,n-1\}$.

For $n=4$, $p_Y[4-k]$ is nonzero for $k \in \{-2,\dots,3\}$. It overlaps $p_X[k]$ at $k=1,2,3$, so $p_Z[4] = \frac{3}{36}$. These three values of k correspond to the rolls $(1,3)$, $(2,2)$, and $(3,1)$.



Counting the overlap for each n gives

$$p_Z[n] = \begin{cases} \frac{n-1}{36}, & 2 \le n \le 7, \\ \frac{13-n}{36}, & 8 \le n \le 12, \\ 0, & \text{otherwise.} \end{cases}$$



(d) Use the flip-and-shift picture to explain why 7 is the most likely sum. Why is $p_Z[n]$ nonzero for exactly 11 values of n? As a check, verify that $\sum_n p_Z[n] = 1$.

Answer: The flipped pulse $p_Y[n-k]$ completely overlaps $p_X[k]$ only when the two pulses line up, which happens at $n=7$. All six values of k then contribute, corresponding to the rolls $(1,6)$, $(2,5)$, …, $(6,1)$. As n moves away from 7 in either direction, the overlap shrinks by one sample per step, which gives the triangular shape.

The overlap is nonempty from $n = 1+1 = 2$ (the smallest sum) to $n = 6+6 = 12$ (the largest sum). This range contains $6+6-1 = 11$ integers, matching the rule that convolving signals of lengths L_1 and L_2 produces a signal of length $L_1 + L_2 - 1$.

Summing the PMF,

$$\sum_n p_Z[n] = \frac{1+2+3+4+5+6+5+4+3+2+1}{36} = \frac{36}{36} = 1.$$